Lab 3: Hero VFX Challenge Documentation

Hero Event: Energy Shield Hit / Break

Overview

For this lab, I created a hero energy shield break effect. The effect is built around a sci-fi shield that starts as a stable glowing surface, then begins to crack, shake, and finally break apart with sparks spawning specifically from the crack lines. I used two VFX Graph systems and one Shader Graph material to build the full effect.

The first VFX system is the main shield effect, which uses a Shader Graph material to show the shield surface and transition it from a normal clean shield into a cracked version. The second VFX system is the spark effect, which appears only when the shield breaks and is controlled so that the particles spawn along the crack pattern instead of across the whole object. Together, these graphs create a layered hero effect with a readable main form, supporting break detail, and a stronger final impact.

1. Intended Mood

The intended mood of this effect is tense, unstable, and high-energy. I wanted the shield to feel like a futuristic protective barrier that is failing under pressure. At first, the shield looks controlled and artificial, but as the crack pattern becomes more visible and the camera shake builds up, the mood changes from stable defense to imminent collapse. The final burst with sparks is meant to feel sudden and dangerous, like the shield can no longer contain the impact and gives way.

The effect was designed to feel cinematic rather than subtle. Because this is a hero VFX moment, the shield needed to be readable immediately and also feel strong enough to support a close-up shot. The build-up before the fracture helps sell anticipation, while the sparks make the final break feel more believable and more energetic.

2. Palette Intention

The main palette of the effect uses cool cyan / blue tones for the shield and warmer yellow/orange spark tones for the break. I chose this contrast on purpose.

The blue and cyan colors make the shield feel technological, clean, and artificial, which fits a sci-fi energy barrier. These colors also work well with emissive materials and bloom, so the shield stays readable even in a darker environment. For the breaking moment, I used warmer

spark colors because they separate the damage event from the base shield. This creates a stronger visual contrast: the shield itself reads as controlled energy, while the sparks read as unstable damage and release of force.

This palette choice also improves readability. If both the shield and the break particles were the same color, the effect would flatten visually. By separating cool shield color from warm fracture sparks, the viewer can instantly tell what is the main surface and what is the damage response.

3. Effect Breakdown / Layer Breakdown

Primary Layer – Normal Shield

The primary layer is the normal shield state, which is the clean and stable version of the shield before any damage appears. This is the most readable part of the effect and clearly shows the main form of the shield to the viewer. It uses a Shader Graph material with emissive color, animated noise, and dot/cell patterning to make the surface feel like energy instead of a solid object.

This layer is important because it establishes the baseline of the effect. When the viewer looks at it, they can immediately understand that it is a shield. The added noise and pattern detail prevent the surface from looking flat and help give it a more dynamic, sci-fi feel. Even if the animation is paused, this layer alone is enough to communicate the purpose of the effect.

Secondary Layer – Crack Transition Layer

The secondary layer is the transition between the normal shield and the cracked shield. This is where the shield starts to visually break down. Instead of instantly switching states, the shader blends between the clean shield and the cracked version using a crack mask and noise.

This layer is what creates the feeling of tension and buildup. The cracks slowly appear and spread across the surface, making the shield look unstable. Because the cracks are driven by textures and masks, the damage looks more natural and controlled instead of random.

This layer connects the beginning and the end of the effect. Without it, the shield would just turn off suddenly, which would feel unrealistic. With the transition, the viewer can clearly see the shield going from stable to damaged before it finally breaks.

Tertiary Layer – Particle Sparks (Break Effect)

The tertiary layer is the particle system that appears when the shield breaks, mainly consisting of sparks. This layer adds motion, impact, and energy to the final moment of the effect.

The important part of this system is that the particles are not spawned randomly. They are generated based on the crack areas, which makes them feel directly connected to the damage on the shield. This makes the effect more believable because the sparks are coming from where the shield is actually breaking.

This layer enhances the final fracture by adding fast, chaotic motion that contrasts with the smoother shield surface. While the shield itself is more controlled, the particles are sharp and energetic, which helps emphasize the break. This makes the final moment feel more impactful and visually interesting.

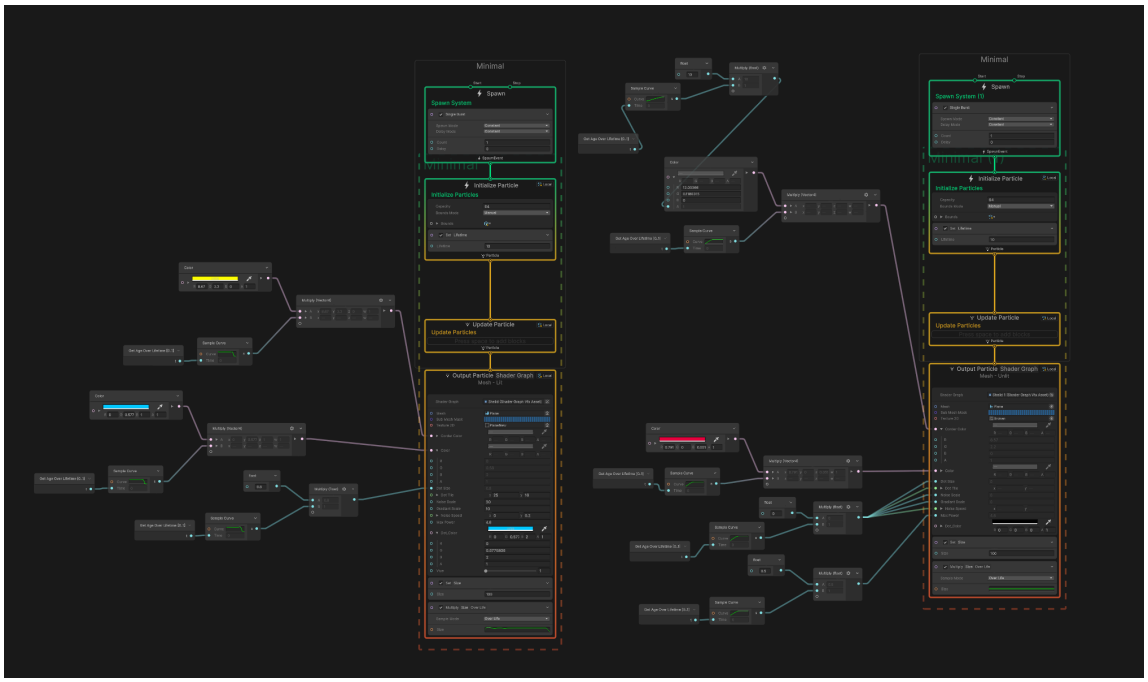
4. VFX Graph Systems Used

VFX Graph 1 – Shield Graph

The first VFX Graph is the main shield graph. This system is responsible for rendering the shield and displaying the shader-based transition between the clean shield and the cracked state.

This graph uses the Shader Graph material as its output, so the VFX Graph is not just spawning a simple particle. Instead, it is using a custom material that controls the look of the shield through emissive color, surface patterning, noise, and crack blending. This lets the shield act as the readable main form of the whole hero effect.

The shield graph is used as the main visual stage of the effect. It exists before the break and carries most of the readability, while also visually preparing the viewer for the fracture event.

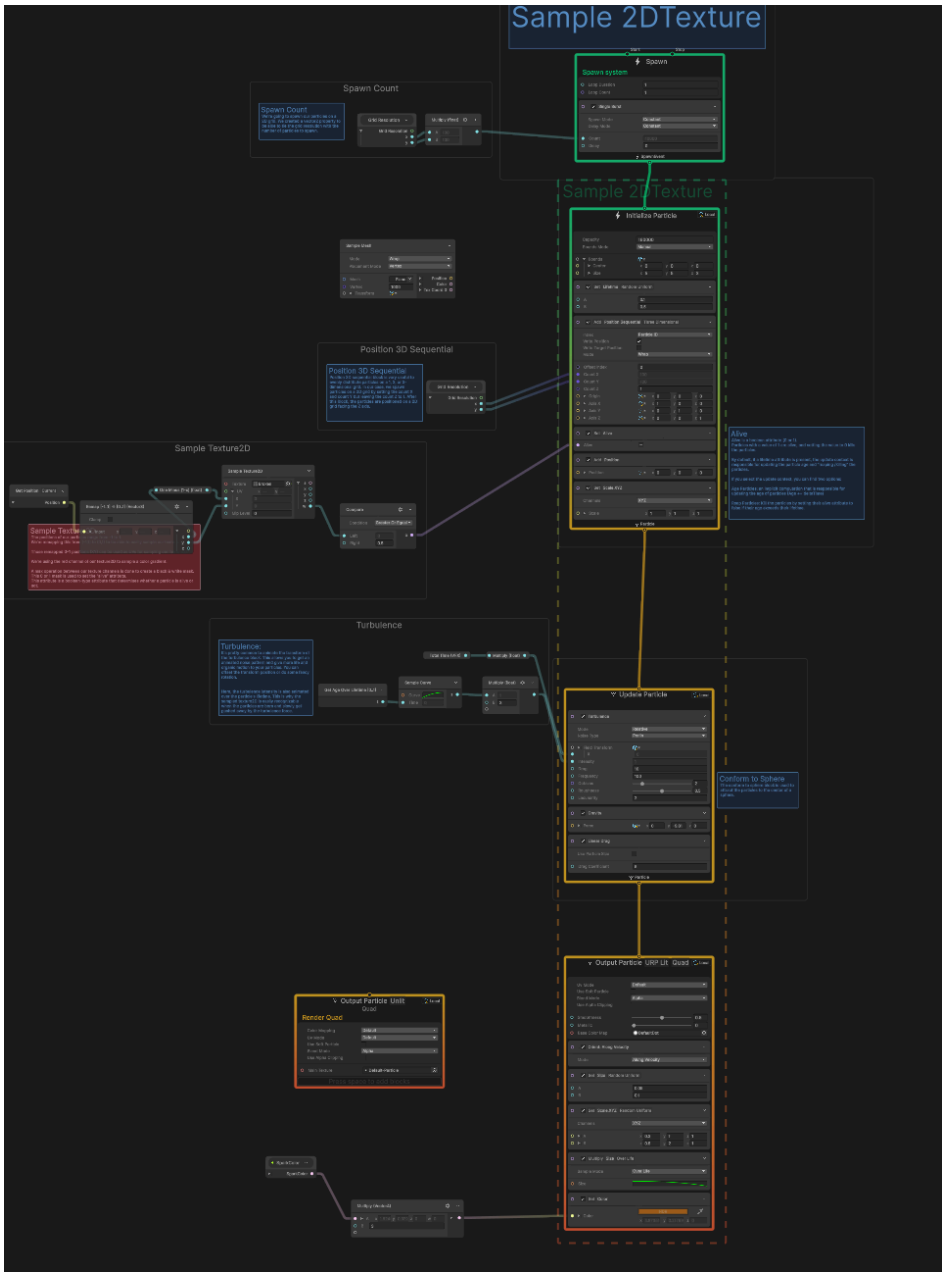


VFX Graph 2 – Spark / Crack Spawn Graph

The second VFX Graph is the spark graph. This graph creates the small spark particles that appear during the break. The important part of this system is that the particles are sampled and spawned based on the crack texture area, so the sparks feel attached to the actual damaged regions of the shield.

This makes the spark system much stronger than if it were just a random burst. It visually ties the secondary layer to the primary layer. The shield's damaged texture is not just cosmetic; it also drives where the particle effect happens. That connection helps the whole effect feel more unified.

This graph acts as the high-energy response layer of the effect. Once the shield reaches failure, the sparks make the break feel more impactful and readable.



5. Shader Graph Breakdown

The Shader Graph is a major part of this effect because it controls the visual language of the shield. The graph combines several elements:

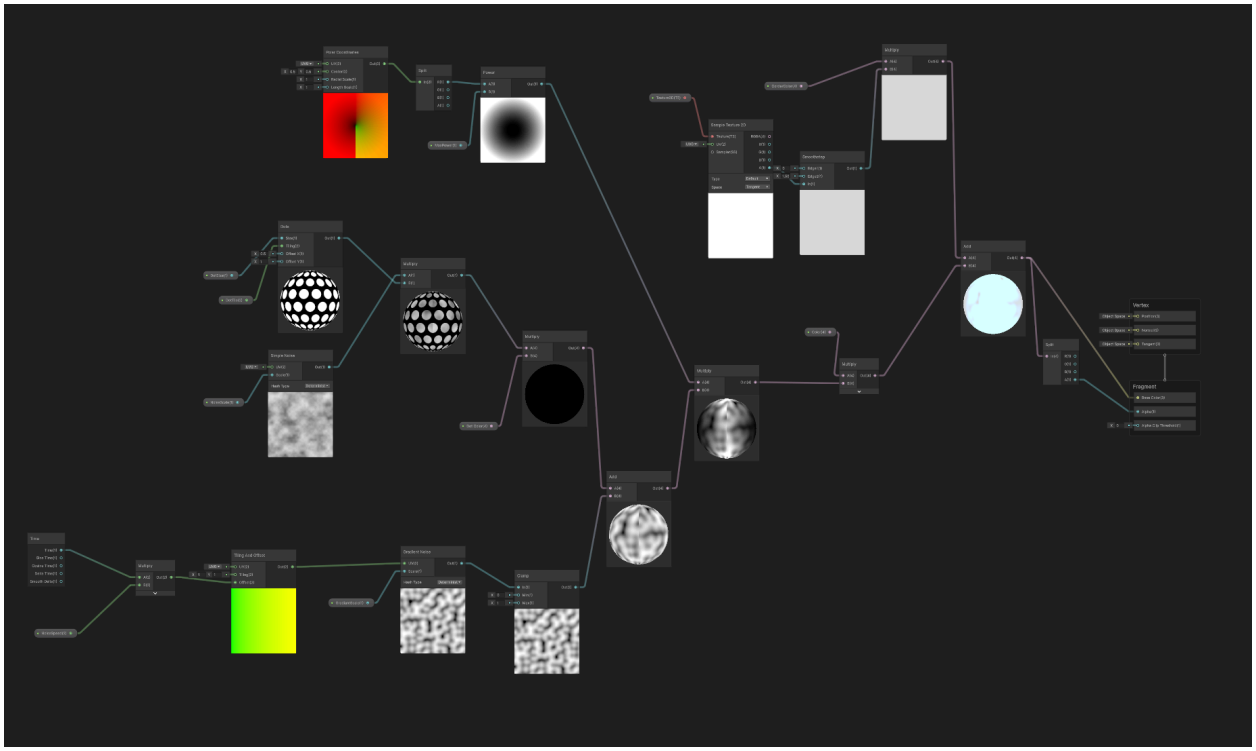
- Base shield color and emissive glow
- Dot / pattern-based detail
- Animated noise
- Crack texture / mask blending
- Normal shield to damaged shield transition

The shader is designed so the shield does not look like a plain transparent sphere. Instead, it feels like a technological energy surface. The dot/pattern detail gives the shield structure, while the noise keeps it from looking static. The crack texture is then used to shift the shield from its normal form into the damaged form.

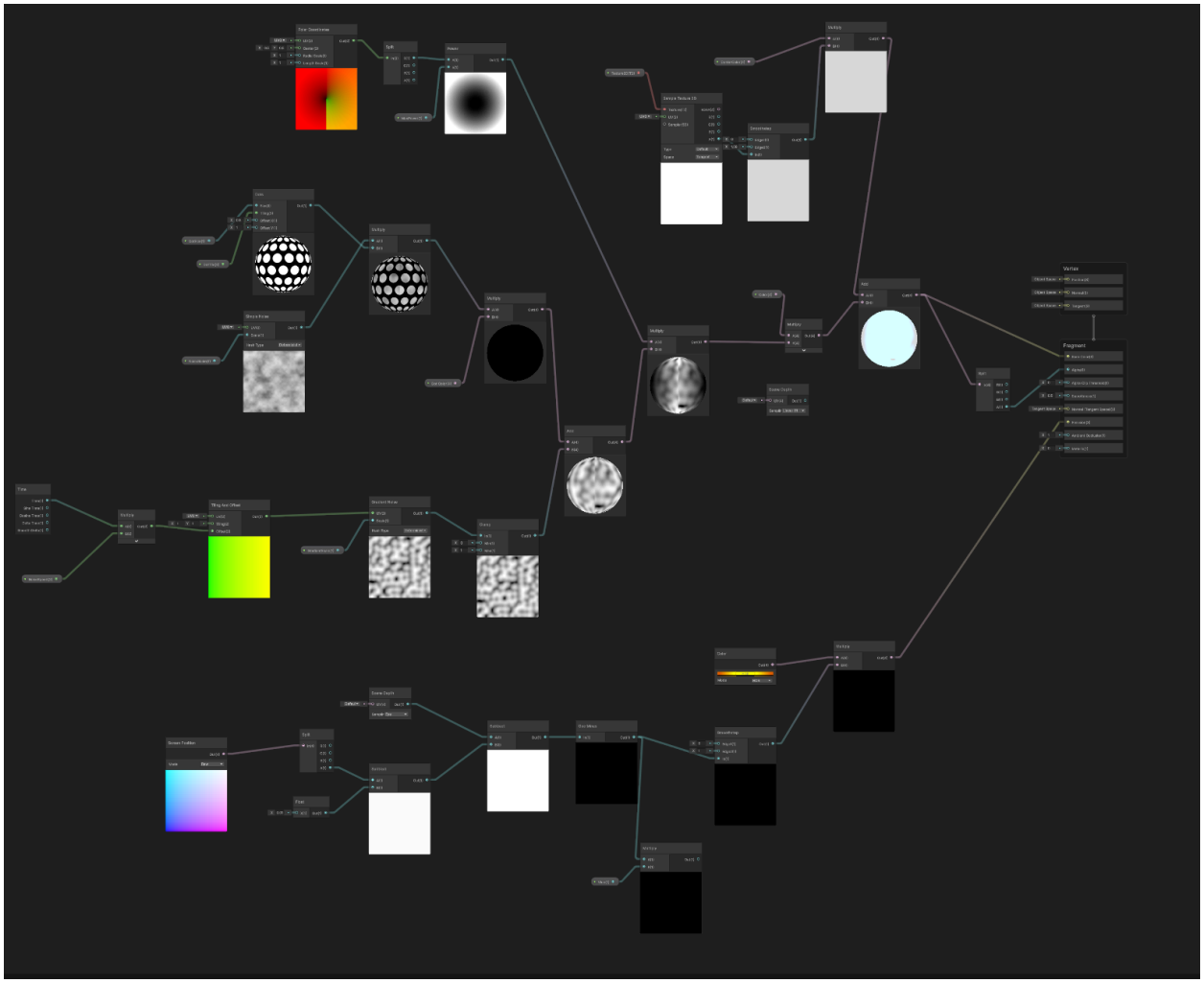
The most important thing about this shader is that it supports storytelling in the effect. It does not just make the shield look pretty; it shows the state change of the shield. The viewer can tell when the shield is stable and when it is beginning to fail.

Because the shader is used inside the shield VFX Graph, it becomes the visual core of the hero event. The spark graph then builds on top of that by reacting to the crack areas.

Cracked Sheild



Normal Sheild



6. Use of Motion Control / Direction

This effect satisfies the requirement of using controlled motion because the break particles are not behaving randomly without context. The spark system is driven by the crack distribution and follows the damaged region of the effect. This means the motion is being controlled by the visual structure of the shield rather than being placed arbitrarily.

There is also a sense of radial and unstable motion in the fracture moment. The final result feels directional because the break event originates from the damaged shield state and is visually reinforced by the spark emission pattern and burst timing.

7. Camera / Framing / Presentation

The effect was built to be shown in a shot where the shield is clearly readable and framed as the focus of the moment. This is important because hero VFX should not feel like background decoration. The camera angle needs to let the viewer see both the shield silhouette and the crack build-up before the break.

The build-up phase works best in a medium-close framing where the audience can see the crack detail and the emission clearly. The break moment becomes stronger when the shot holds just long enough for the viewer to register the instability before the sparks appear.
Post-processing such as bloom helps the emissive shield read more strongly, especially against a darker environment.

8. Optimization

What I Optimized

For optimization, I focused on particle count / spawn control and keeping the secondary spark system limited to crack areas only instead of spawning particles across the full shield surface.

This is an important optimization because if the sparks were allowed to spawn everywhere, the effect would create more particles than necessary and also increase visual noise. By restricting particle emission to the crack mask, I improved both performance and readability at the same time. Fewer particles are wasted in non-important areas, and the particles that do appear are more meaningful.

I also kept the effect visually focused so that the primary shield remains readable. This avoids a situation where too many particles or too much overdraw hides the main form.

Before Optimization

Before optimization, the secondary break particles were less controlled and could cover a wider area than needed. This increased visual clutter and made the final fracture less focused. From a performance point of view, it also meant more work was being done for particles that were not helping the readability of the effect.

In the “before” version, the effect had the risk of becoming noisier and more expensive than necessary because the break detail was not as selective.

After Optimization

After optimization, the spark system was constrained to the crack regions, which reduced unnecessary particle spawning. This made the effect cleaner, more intentional, and easier to

read. The break now looks like it is coming from actual damage points instead of appearing as a generic particle burst.

This change is measurable because it reduces the number of particles being used in non-critical areas and makes the system more focused. It also supports the artistic goal: the viewer's eye is directed to the crack lines, which are the most important feature of the break.

Why This Optimization Was Chosen

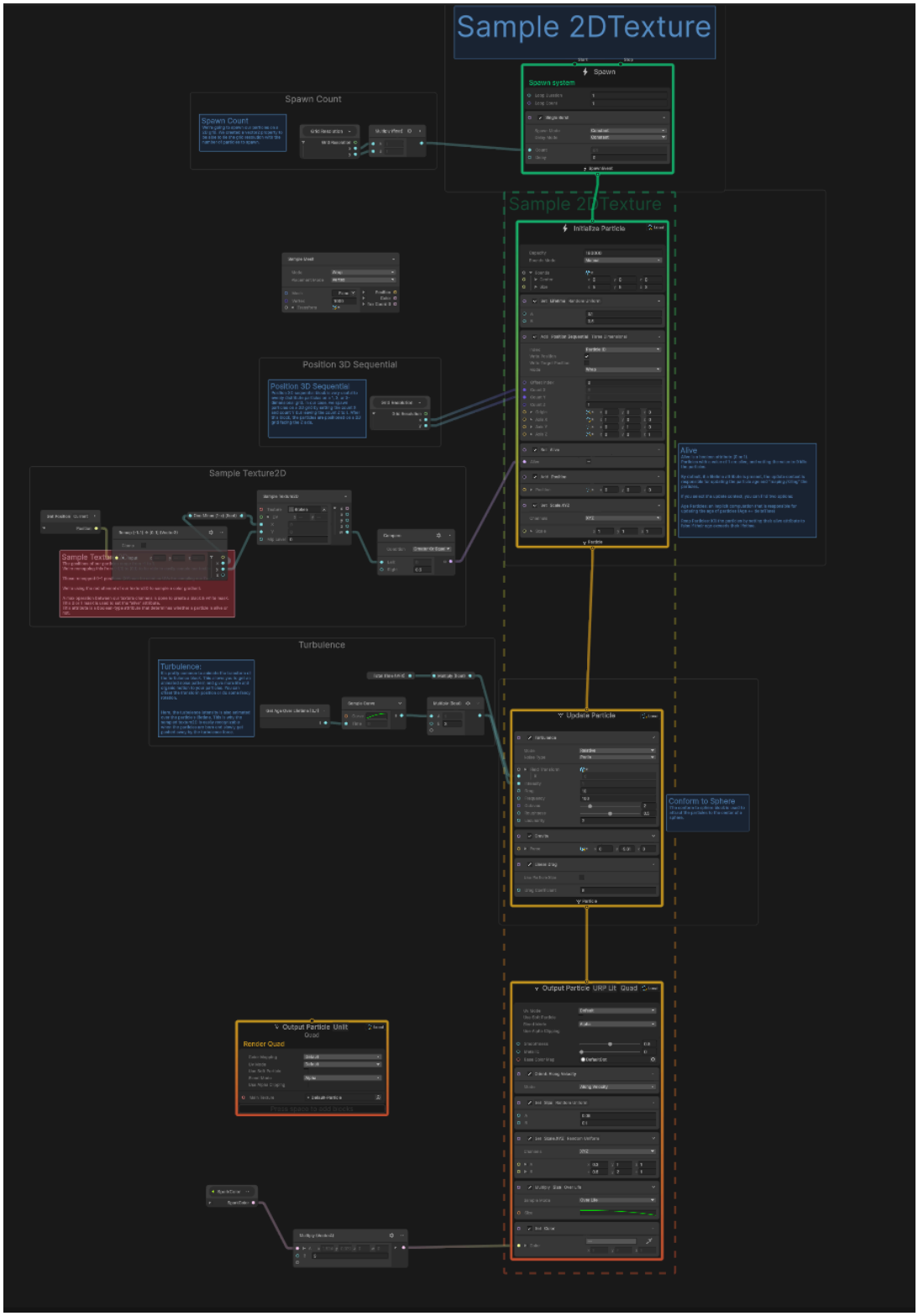
I chose this optimization because it improved both technical efficiency and visual clarity without damaging the look of the effect. Some optimizations can make an effect look noticeably worse, but in this case the change actually made the effect stronger. The sparks now reinforce the crack pattern instead of competing with the main shield.

This was a good optimization choice for a hero VFX because hero effects need to remain readable first. Reducing unnecessary particles helped preserve that readability.

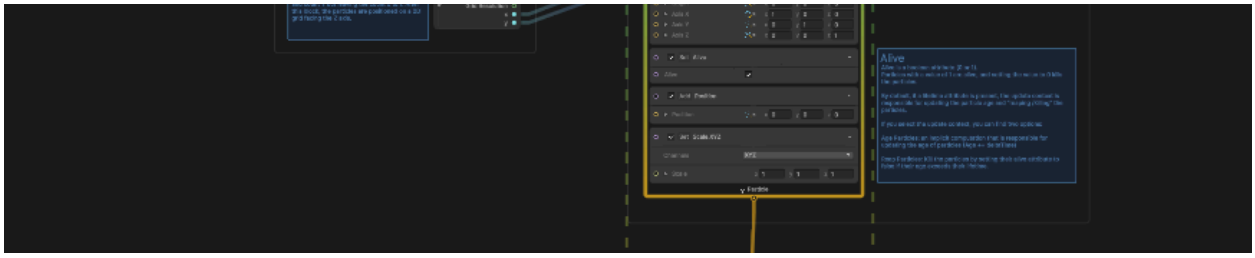
9. Proof of Improvement

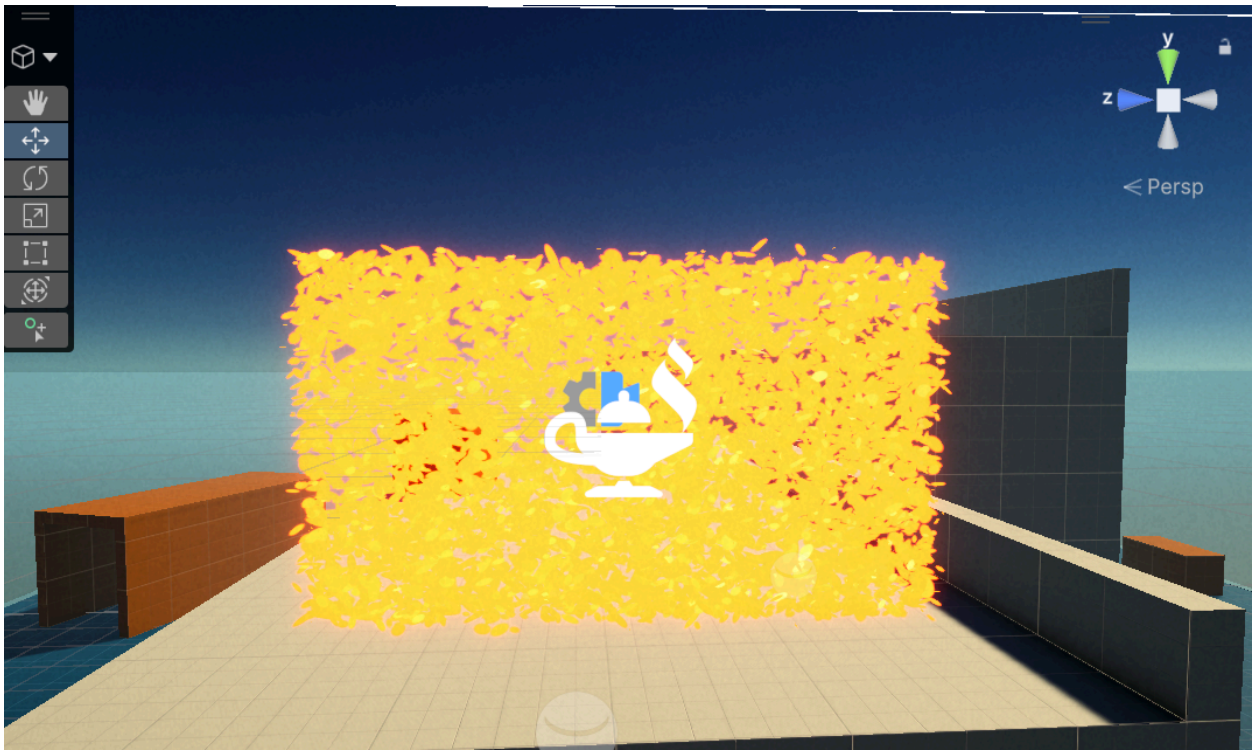
For proof of improvement, I included:

1. Graph overview

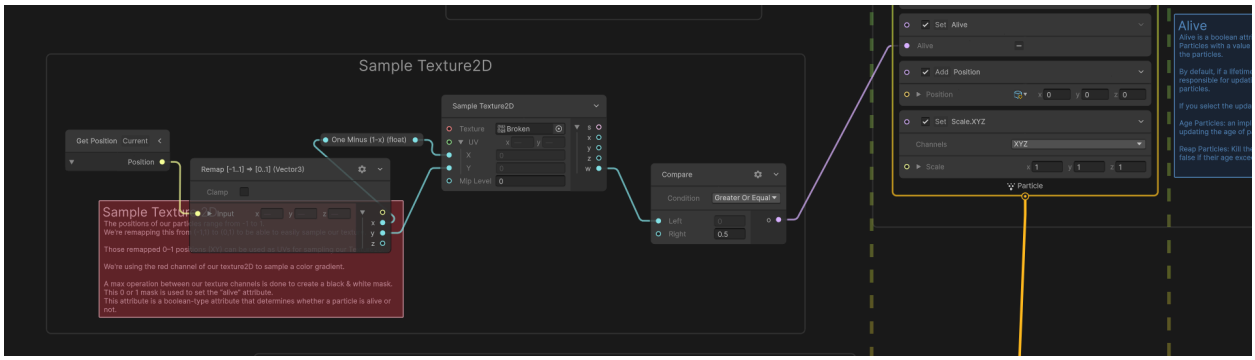

Sample 2DTexture
Spawn
Spawn system
Spawn Count
Sample 2DTexture
Initialize Particle
Position 3D Sequential
Sample Texture2D
Turbulence
Update Particle
Conform to Sphere
Output Particle URP Lit Quad
Output Particle Unlit Quad
Render Quad

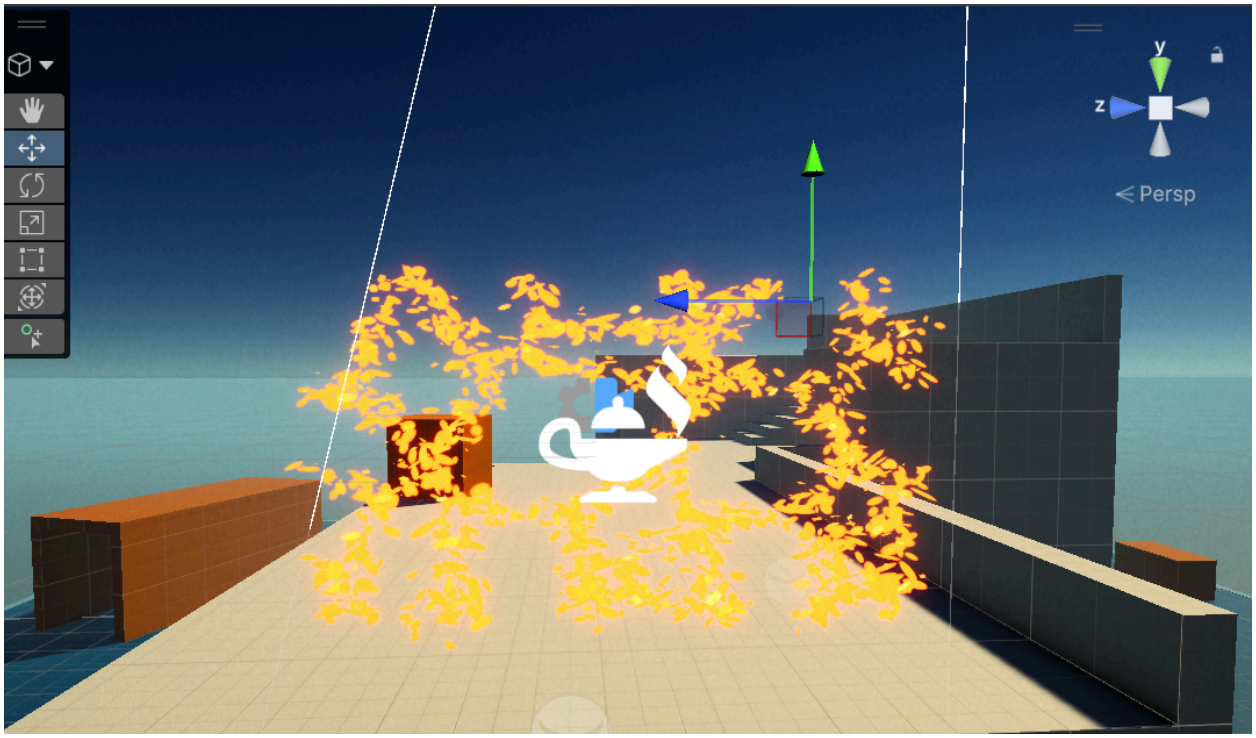
2. Before optimization screenshot





3. After optimization





These screenshots demonstrate that the optimized version is both cleaner visually and more efficient because particle use is concentrated only where the effect needs it.

10. What Worked Well

The strongest part of this effect is the relationship between the shield shader and the spark graph. The main shield is readable as the primary layer, and the sparks only appearing on the crack areas makes the secondary layer feel connected instead of random.

Another part that worked well is the color contrast. The cool shield and warm spark colors make the effect easier to understand immediately. The transition from stable shield to crack state also gives the effect a stronger sense of progression, which helps it feel like a hero moment instead of a basic particle event.

11. What I Would Improve Next

If I continued developing this effect, I would add a stronger tertiary lingering layer, such as faint embers, residual energy haze, or a short-lived distortion pass after the shield breaks. That would make the effect feel even more complete and would strengthen the after-impact moment.

I would also experiment further with camera timing and environmental lighting so that the shield break feels even more cinematic in context, not only in isolation.

12. Conclusion

Overall, this effect meets the lab requirements by using multiple VFX Graph systems, a Shader Graph material, a clear primary / secondary / tertiary layer structure, and a measurable optimization pass. The final result is a sci-fi energy shield break effect that is readable, layered, and cinematic.

The primary shield surface communicates the effect clearly, the crack transition adds tension, and the spark system gives the fracture moment energy and impact. The optimization improved the effect by reducing unnecessary particle spawning and making the final result more focused on the crack areas. This made the effect not only more efficient, but also visually stronger.

References -

https://www.youtube.com/watch?v=m7GimQakt3A&t=211s

https://www.youtube.com/watch?v=0SsJlS-Zsmg

https://www.youtube.com/watch?v=rB4YMQmO8Mw&t=1007s